

## Kelly Buabeng

Accra, Ghana | 0268668832 | kbbuabeng002@st.ug.edu.gh

### Personal Summary

Results-driven **backend software engineering and IT professional** with hands-on experience building, optimizing, and supporting backend systems across startup and enterprise environments. Demonstrated expertise in backend development through practical experience at **Xavs Labs**, where I contributed to API implementation, database optimization, code reviews, and technical documentation for web applications. Proven ability to support and enhance enterprise IT infrastructure within large organizations, including **TotalEnergies Marketing Ghana PLC** and the **Ghana Civil Aviation Authority**, combining software engineering skills with strong system administration and troubleshooting capabilities. Possesses a solid foundation in applying cybersecurity best practices to safeguard systems, data, and network access. Backed by a strong academic background in Mathematical Sciences and a forward-thinking approach to building secure, scalable, and maintainable backend solutions.

### Skills

- Network Security
- Database Management
- Software Development
- Technical Support & Troubleshooting
- System Optimization
- Data Analysis
- IT Documentation
- Project Management
- Collaboration & Multitasking
- Research Methodologies

### Technical Stack

- REST APIs

- Git & GitHub
- Python
- CLI Tools
- Vercel Deployment
- Prolog

## Work Experience

### System Administrator / IT Technician (Internship)

#### **Total Energies Marketing Ghana PLC — Accra, Ghana**

March 2024 – June 2024

- Manage and maintain IT systems, troubleshoot hardware/software issues, and ensure network security.
- Install and configure software, provide technical support, and optimize system performance.
- Involved in database management, cybersecurity practices, and backend development to enhance IT infrastructure.
- Collaborate to implement innovative solutions, improve efficiency, and ensure seamless technology operations.

### Software Developer / Research Analyst (Internship)

#### **Zormor — Accra, Ghana**

November 2023 – March 2024

- Developed and maintained software applications while conducting research to improve system performance and security.
- Responsible for coding, troubleshooting, and analyzing data to support software development.
- Researched emerging technologies and industry trends to enhance efficiency and innovation.
- Worked collaboratively to solve technical challenges and improve software functionality.

## Backend Developer (Intern)

### **Xavs Labs — Accra, Ghana**

August 2025 – January 2026

- Assisted in backend development for web applications using modern frameworks.
- Implemented APIs and optimized database queries for performance.
- Collaborated with senior developers to troubleshoot and resolve technical issues.
- Contributed to code reviews and documentation for development processes.

## IT & Cybersecurity Intern

### **Ghana Civil Aviation Authority — Accra, Ghana**

November 2025 – December 2025

- Supported IT infrastructure maintenance and cybersecurity monitoring.
- Assisted in implementing security protocols to safeguard sensitive data.
- Participated in vulnerability assessments and risk analysis.
- Provided technical support and contributed to IT documentation.

## Education

BSc Mathematical Science (Computer Science with Statistics)— University of Ghana, Accra (2022 - 2026)

WASSCE General Science — Presbyterian Boys Secondary School (PRESEC-LEGON), Accra (2019 - 2022)

## Selected Projects

- Exoplanet Classification ML Model: Machine learning model using NASA's Kepler dataset to identify and classify exoplanets with advanced data processing techniques.
- Expert System for Illegal Mining in Ghana: A generative AI logic system using Python Flask and Prolog to assess pollution risks from illegal mining in Ghana.
- Product Z: An AI system to help African SMEs as a smart business assistant (Ongoing).
- Chatbot Application: Real-time messaging service with Node.js backend and React frontend.

- NHIS Redesign Project: UX redesign with React/Next.js.

## Achievements & Certifications

- **ALX Professional Foundations Certificate**  
( <https://savanna.alxafrica.com/certificates/Nc6Fhpy7e> )— Completed intensive professional training focused on work ethics, communication, collaboration, leadership, and problem-solving in professional environments.
- **NASA Space Apps Challenge 2025 (Top 2 Team, Ghana)** — Led development of a machine learning solution using NASA's Kepler exoplanet dataset; ranked top 2 nationally and selected to represent Ghana for delivering a high-impact, data-driven space science solution.

## Languages

English (Fluent)

Twɩ and Fante (Fluent)

## References

Emmanuel Kojo Awortwi | +233 50 675 8109 | [emmanuel.awortwi@totalenergies.com](mailto:emmanuel.awortwi@totalenergies.com) |  
Total Energies Marketing Ghana PLC

Dorothy | +233 50 006 7398|Ghana Civil Aviation Authority

Denzel | +233 54 395 1760 | XavsLabs